

Artificial Intelligence Lab (AI1002)

Date: March 19th 2025

Course Instructor(s)

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Lab Mid Exam B

Total Time: 90 minutes

Total Points: 50

Total Questions: 03

Semester: Spring 2025

Campus: Karachi

Department: CS, AI, SE, CY &DS

Instructions:

- Submit a .py file or .ipynb file for all questions.
- Each file must contain a header with your roll number.
- Also submit a ".pdf" OR ".docx" file containing screenshots of codes and clearly show the output.
- All your codes must produce output.
- For Question 03, the code is available on local storage (\\172.16.5.43\Exam Material\ (Your Teacher Name)).

Student Name

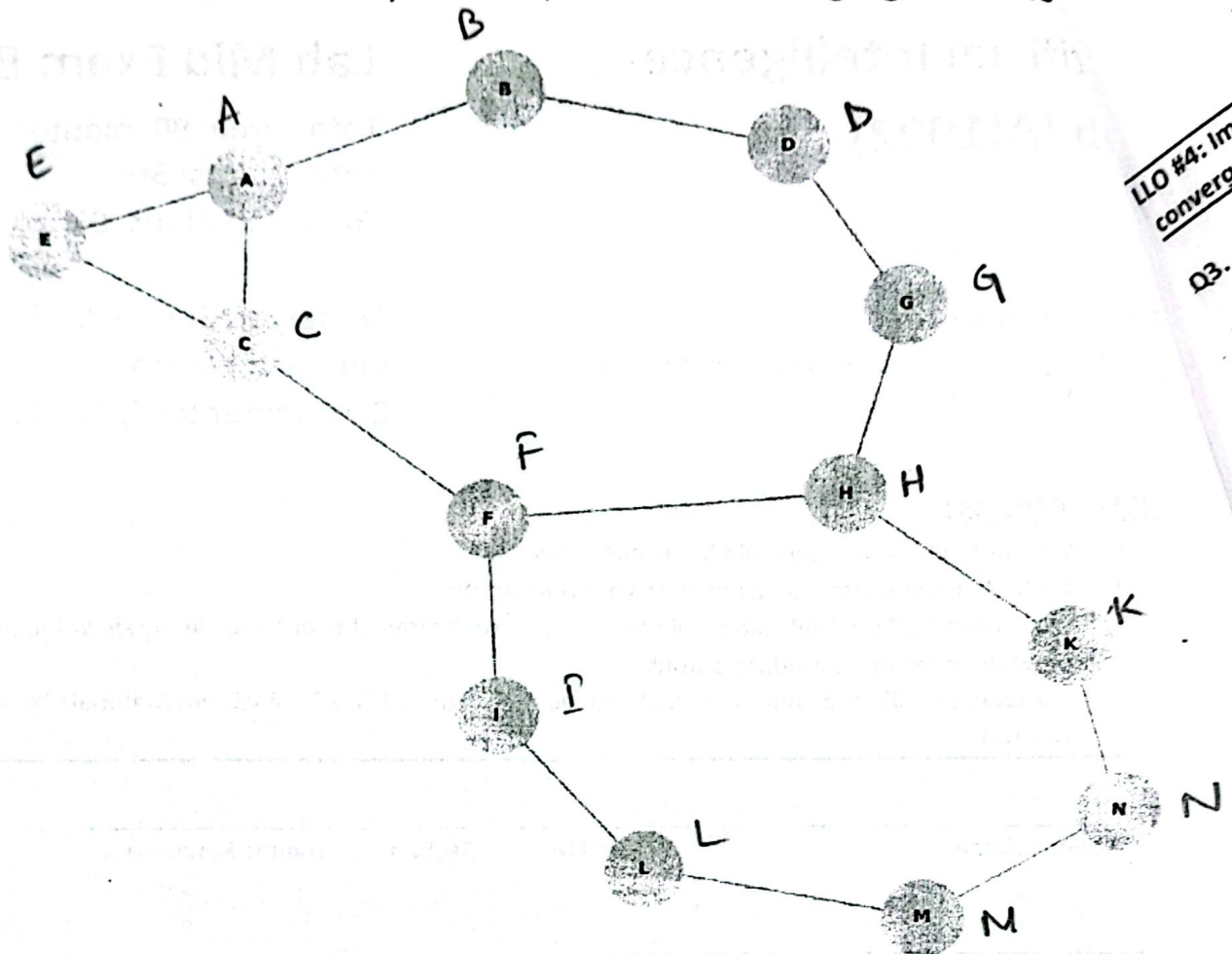
Roll No

Section

Student Signature

LLO # 1: Explain and implement BFS and DFS algorithms, comparing their efficiency and applications in uninformed search. [18 points]

Q1 You are developing a goal-based robot exploration agent that needs to navigate through a graph. The graph is represented as an undirected graph, where each node represents a room, and the edges represent the doors connecting these rooms. The robot's goal is to navigate from a starting room (E) to a target room (N). You need to use Depth-First Search (DFS) to solve this.



LLO #4: Implement G
convergence, and

Q3.

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LLO #3: Apply and implement A* and GBFS algorithms, evaluating their heuristics, efficiency, and applications in informed search. [18 points]

Q2. A drone is deployed in a 5×5 disaster relief zone to deliver emergency supplies to a survivor. The drone starts at (0,0) and must reach the survivor's location at (3,4) as quickly as possible using A search algorithm* with Manhattan distance as the heuristic function.

The disaster zone contains debris (represented as 1s in the grid), which the drone cannot fly through. The drone can move up, down, left, or right, but not diagonally.

Grid Representation:

```
grid = [
  [0, 0, 0, 0, 0],
  [0, 1, 1, 1, 0],
  [0, 0, 0, 1, 0],
  [0, 0, 0, 1, 0],
  [0, 1, 0, 0, 0],
]
```

LLO #4: Implement Genetic Algorithms to solve optimization problems, analyzing their operators, convergence, and performance. **[14 points]**

Q3.

Instructions:

The code is available on local storage (\\172.16.5.43\Exam Material\ (Your Teacher Name)).

Problem Statement:

A delivery company has several delivery trucks and needs to find the optimal routes for each truck to deliver packages to multiple locations. Each truck has a limited capacity and each delivery point has a specific demand. The goal is to minimize the total travel distance while ensuring that each truck can carry the required amount of packages and that no truck exceeds its capacity.

Your task is to implement a Genetic Algorithm (GA) that will optimize the delivery routes for the trucks, considering the capacity constraints, delivery demands, and distance between delivery points.

Task:

1. By using Tournament Selection method (individual with the best fitness in this subset is chosen for reproduction), create a selection function. [4]
2. You are expected to write a crossover function using Two-Point crossover. [4]
3. You are expected to write a mutate function using Uniform Mutation (each gene in an individual has an equal probability of being randomly altered). [4]
4. Update such that the code runs completely and outputs best solution. [2]